

# HPE CPP

## Week-4 Report

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# QUIC Traffic Classification Using Observable Metadata

A Machine Learning-Based Approach for Encrypted Network Traffic Analysis

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## 1. Overview of Dataset and Feature Engineering

### Traffic Capture and Flow Construction

The dataset was constructed from packet-level captures of live QUIC traffic generated by seven real-world application categories: video streaming, live streaming, short-form video, file download, web browsing, chat, and VoIP. Raw captures were collected in PCAP format and processed into flow records using a time-window segmentation strategy. A flow corresponds to a continuous segment of a QUIC connection, identified by its Connection ID, and is described entirely by statistical features computed over the packets within that window.

To maximize sample volume and ensure that transient traffic behaviours are adequately represented, a sliding window approach was applied: each session was divided into 10-second windows with a 5-second overlap between consecutive segments. This strategy captures behavioral transitions such as the shift from a content-loading burst to a steady-state delivery phase as distinct labeled data points, enriching the training set with the full behavioral range of each traffic class.

The final dataset contains 296 flow records across seven classes. No payload content was accessed at any stage, in full compliance with the encryption. The class distribution is notably uneven: the video class contains 109 records, live streaming 64, chat 50, voip 28, short\_video 25, download 13, and browsing 7. This imbalance has direct consequences for model training and evaluation, discussed further in subsequent sections.

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### Feature Categories and Their Relevance

#### Packet Size Features

Mean packet size (`avg_pkt_size`) reflects the dominant transfer mode of a flow. High-throughput delivery classes such as video and download exhibit large mean sizes approaching the QUIC path MTU, while low-bandwidth classes such as VoIP and chat produce characteristically small averages. Standard deviation (`std_dev`) captures size variability, distinguishing mixed flows containing both large data frames and small control packets from uniform flows. Maximum and minimum packet sizes (`max_pkt_size`, `min_pkt_size`) provide boundary context: minimum values often correspond to ACK, PING, or connection management packets, and are particularly diagnostic for chat and VoIP traffic. The interquartile range (`size_iqr`) offers a robust variability measure less sensitive to outliers than standard deviation, providing a complementary view of packet size consistency across the window.

## Timing Features -Inter-Arrival Time (IAT)

Mean inter-arrival time (`avg_iat`) measures the average gap between consecutive packet arrivals. Video and download flows exhibit sub-millisecond average IAT reflecting continuous high-throughput delivery, while chat and VoIP flows show larger, more periodic gaps. IAT standard deviation (`iat_std`) captures delivery regularity: adaptive bitrate streaming classes such as `live` and `short_video` exhibit high IAT variability as they alternate between rapid burst delivery and idle gaps, whereas steady-state flows produce low variability. Together, these two features encode the temporal rhythm that distinguishes application behaviour at the flow level.

## Volume and Rate Features

Packet count (`pkt_count`) and flow duration (`duration`) establish the scale of each window. Packet rate (`pkt_rate`), computed as packets per second, summarizes throughput intensity and is among the most discriminative features for separating high-demand media classes from low-activity traffic such as browsing or chat.

## Burstiness

The burstiness metric quantifies the degree to which packet arrivals cluster in time rather than distributing evenly. High burstiness characterizes streaming protocols that prefetch content in rapid bursts followed by silence, most evident in video, `short_video`, and download, while interactive classes such as chat and VoIP exhibit low, rhythmic burstiness. This feature captures higher-order behavioural structure not derivable from any single size or timing statistic.

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## Effectiveness Despite QUIC Encryption

QUIC encrypts its payload and all QUIC-layer header fields using TLS 1.3, rendering deep packet inspection infeasible. However, IP and UDP layer attributes specifically packet lengths and transmission timestamps remain fully observable at the network level regardless of encryption. The premise of this work is that different application categories generate statistically distinguishable traffic patterns at the flow level. A video stream demanding sustained high throughput produces a measurably different size and timing profile than a VoIP session with fixed codec payloads, which differs in turn from a chat session with sporadic short messages. These behavioural fingerprints are determined by application-layer delivery logic and are not altered by the encryption layer, making flow-level statistical features a viable and privacy-respecting basis for traffic classification.

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## 2. Machine Learning Models Used

### Decision Tree

A Decision Tree classifier was included as the interpretable baseline model. It partitions the feature space through recursive binary splits, selecting the feature and threshold at each node that maximizes information gain. The model's primary advantage is transparency: the resulting decision rules can be directly inspected to identify which features and thresholds distinguish traffic classes, which is operationally valuable for network management deployment. Its principal limitation is susceptibility to overfitting, particularly on small or imbalanced datasets. Without depth constraints or pruning, a decision tree can memorize window-specific noise rather than learning generalizable patterns, producing unstable per-class performance. In this study, the browsing and download classes with only 7 and 13 total samples respectively are especially vulnerable to this effect.

Random Forest

Random Forest was selected as the primary model. It constructs an ensemble of decision trees, each trained on a bootstrapped sample of training data with a randomly selected feature subset considered at each split, and determines the final prediction by majority vote. This ensemble mechanism substantially reduces variance compared to a single tree, yielding more stable and generalizable predictions. Random Forest requires no feature normalization, tolerates moderate class imbalance, and naturally provides feature importance scores. In the QUIC traffic context, where the sliding window methodology introduces correlated adjacent samples and different features contribute unevenly across traffic types, the ensemble averaging property is directly beneficial. These properties make Random Forest the most appropriate primary model for this dataset.

XGBoost

XGBoost implements gradient-boosted decision trees, constructing the ensemble sequentially where each new tree corrects the residual errors of the current model using gradient descent. It is widely recognized for strong performance on tabular datasets when sufficient training data and hyperparameter tuning are available. XGBoost was included to assess whether boosting could surpass Random Forest's bagging approach. Boosting methods are more sensitive to dataset scale and class distribution than bagging methods, and their advantage is most pronounced with plentiful, balanced training data. With the current dataset, particularly the underrepresented browsing, download, and short\_video classes, XGBoost's learning capacity was constrained, yielding lower performance than Random Forest without extensive tuning.

3. Model Performance Analysis

Overall Accuracy

Decision Tree and Random Forest both achieved 91.67% overall accuracy on the 60-sample held-out test set, while XGBoost achieved 83.33%. The accuracy parity between Decision Tree and Random Forest is notable but misleading at the aggregate level: the two models differ meaningfully in per-class stability and the distribution of their errors, as the classification reports reveal.

| Model         | Accuracy | Macro Avg F1 | Weighted Avg F1 |
|---------------|----------|--------------|-----------------|
| Decision Tree | 91.67%   | 0.91         | 0.91            |
| Random Forest | 91.67%   | 0.90         | 0.91            |
| XGBoost       | 83.33%   | 0.81         | 0.82            |

Class-Level Performance: Strong Classes

**Class 2 - Download** achieved perfect precision, recall, and F1-score (1.00) across all three models, reflecting the distinctiveness of bulk file transfer flows. Download traffic generates near-maximum packet sizes, very high packet rates, and low IAT variability, forming a tight, well-separated cluster in the feature space that all models identify reliably.

**Class 5 -Video** was the strongest and most consistently classified class across all models, with F1-scores of 1.00 (Decision Tree), 0.98 (Random Forest), and 0.96 (XGBoost). With 21–22 test samples and 109 training records, it is the best-represented class, and its high-throughput, high-burstiness profile makes it statistically distinct.

**Class 0 - Browsing**, despite being the smallest class (support of 1–2 in the test set), achieved perfect scores across all models. While this result should be interpreted cautiously given the minimal test support, it suggests that the browsing traffic captured generates sufficiently distinct short-burst, low-rate patterns to be reliably separated from other classes.

#### **Class-Level Performance: Weak Classes**

**Class 4 - Short Video** is the weakest-performing class across all models. XGBoost achieved an F1-score of only 0.50, with recall of 0.40, meaning three of five test samples were misclassified, primarily as live streaming (class 3). Decision Tree reached 0.86 F1 (recall 0.75), while Random Forest achieved the best result at 0.89 F1 with perfect recall (1.00). The confusion between short\_video and live is structurally motivated: both classes involve adaptive bitrate delivery with bursty IAT patterns and variable packet sizes, making them difficult to distinguish at specific operating points.

**Class 6 - VoIP** showed consistent weakness in recall across all three models, achieving F1-scores of 0.67 (XGBoost), 0.75 (Decision Tree), and 0.67 (Random Forest). With only 5–6 test samples, VoIP flows that deviate from the expected low-size, periodic pattern, such as during codec negotiation or quality shifts, are misclassified as chat (class 1). The behavioral overlap between VoIP and chat during low-activity phases is the primary driver of this confusion.

**Class 1 - Chat** showed moderate performance, with F1-scores of 0.70 (XGBoost), 0.89 (Decision Tree), and 0.85 (Random Forest). XGBoost misclassified 3 of 10 chat samples, two as video (class 5) and one as live (class 3), suggesting that during periods of elevated message activity, chat flows can produce packet rates and sizes that superficially resemble lower-intensity media flows.

#### **Confusion Matrix Interpretation**

Across all three models, misclassification is concentrated along two axes: the short\_video/live boundary and the voip/chat boundary. These are not random errors; they reflect genuine statistical overlap between behaviorally similar classes at specific operational states. The video class (class 5) is essentially confusion-free in all models, with at most one misclassified sample. The download class (class 2) is perfectly classified in every model. The browsing class (class 0), despite minimal test support, is never confused with any other class.

XGBoost's confusion matrix reveals a more dispersed error pattern: it misclassifies live as short\_video (3 samples), voip as chat (2 samples), and chat as both live and video, indicating that without the variance reduction of an ensemble, the model struggles more with borderline samples in underrepresented classes. Decision Tree and Random Forest show tighter, more localized confusion concentrated only at the known class-boundary overlaps.

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## **4. Comparative Insights**

### **Why Random Forest Performed Best**

Random Forest's advantage over the Decision Tree, despite identical overall accuracy, lies in the distribution of its performance. Random Forest achieves a balanced macro-averaged F1 of 0.90 compared to the Decision Tree's 0.91, but this near-parity conceals an important qualitative difference: Random Forest correctly classifies all four short\_video test samples (recall 1.00, F1 0.89), while the Decision Tree misses one. Random Forest also produces zero false positives for the live class (precision 1.00), whereas the Decision Tree generates one. The ensemble mechanism prevents any single poorly-learned split from dominating classification decisions, yielding more reliable boundary

handling for the short\_video/live overlap, the most challenging region of the feature space in this dataset.

### **Decision Tree: Interpretability vs. Stability**

The Decision Tree achieves strong performance on this dataset, particularly for the majority classes with ample training support. Its macro F1 of 0.91 is the highest of the three models. However, this result reflects favorable test set composition more than guaranteed generalization: a single tree trained on a different random split of the small dataset could produce substantially different per-class performance, especially for minority classes. Its primary operational value is interpretability and speed of inference, making it suitable as a diagnostic tool or for deployment contexts where rule transparency is required.

### **XGBoost: Dataset Scale Constraint**

XGBoost's 83.33% accuracy and macro F1 of 0.81 should be understood as a dataset-scale limitation rather than an algorithmic deficiency. Gradient boosting requires sufficient samples per class across ensemble iterations to learn meaningful residual corrections. With browsing having only 7 total samples, short\_video 25, and download 13, XGBoost cannot fully exploit its sequential error-correction mechanism. The per-class results confirm this: XGBoost performs competitively on well-represented classes (video: F1=0.96, live: F1=0.83) but degrades on minority classes (short\_video: F1=0.50, voip: F1=0.67). With a larger, more balanced dataset and systematic hyperparameter tuning, XGBoost would be expected to match or surpass Random Forest.

### **The Importance of Precision, Recall, and F1**

Accuracy alone is insufficient for evaluating classifiers on imbalanced datasets. The video class constitutes approximately 37% of the dataset; a model predicting video for every sample would achieve misleadingly high accuracy. Examining per-class precision and recall reveals where each model genuinely succeeds and where it fails. For instance, Random Forest's precision of 1.00 for live traffic means it never incorrectly assigns a sample to that class, while its recall of 0.92 indicates one missed live sample, a meaningfully different failure mode than XGBoost's recall of 0.92 for live but precision of 0.75, which means it misassigns non-live samples to that class. These distinctions have direct operational consequences in network traffic management, where false positives and false negatives carry different costs depending on the application.

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## **5. Key Observations**

### **Feature Engineering Drives Discriminative Power**

The eleven features collectively provide orthogonal views of flow behavior that no single feature could supply alone. Packet rate and mean packet size jointly identify high-throughput classes, but they are insufficient alone to distinguish video from download. IAT standard deviation and burstiness together reveal the difference between smooth high-rate delivery (download) and burst-pause streaming (video, short\_video, live). The size IQR separates chat and VoIP — both low-mean-size classes — by capturing the packet size consistency that distinguishes fixed-codec VoIP payloads from the variable-length messages of chat applications. This multi-dimensional feature design is what enables classes with similar mean behavior to be separated by their variability structure.

### **Role of Overlapping Windows**

The 5-second stride with 10-second windows increased the effective sample count from what would have been achievable with non-overlapping segmentation, particularly for shorter capture sessions. More importantly, overlapping windows ensure that behavioral transitions — adaptive rate changes,

buffering events, connection renegotiations — are sampled multiple times, exposing models to the range of states within each class. The trade-off is inter-sample correlation: adjacent windows share half their packets, meaning their feature vectors are not statistically independent. This requires that train-test splits respect session boundaries rather than randomly assigning individual windows, to avoid inflated performance estimates from near-duplicate records appearing in both sets.

## QUIC Encryption Does Not Prevent Traffic Classification

The central finding of this study is that flow-level metadata derived exclusively from observable IP/UDP attributes, packet sizes and transmission timestamps, is sufficient to classify seven categories of encrypted QUIC traffic with over 91% accuracy. QUIC's TLS 1.3 encryption, while robust against payload inspection, does not suppress the behavioral side-channel information that applications embed in their traffic patterns through delivery logic, codec characteristics, and session management behavior. The burstiness of video prefetching, the periodicity of VoIP codec frames, the short-burst signature of chat messages, and the sustained throughput of file downloads are all preserved as observable flow-level patterns regardless of payload encryption. This demonstrates that encrypted traffic classification is feasible at production accuracy using only legally and technically observable network metadata.

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## 6. Limitations

**Class imbalance and small sample sizes.** The dataset is significantly imbalanced: video (109 samples) and live (64 samples) dominate, while browsing (7 samples), download (13 samples), and short\_video (25 samples) are substantially underrepresented. This imbalance affects both training, where minority classes have fewer examples to generalize from, and evaluation, where test-set support for minority classes is too small to produce statistically stable performance estimates. The browsing class, with a test support of 1–2 samples, produces a precision and recall of 1.00 that cannot be reliably interpreted.

**Overlapping behavioral profiles.** The short\_video/live boundary and the voip/chat boundary represent genuine statistical ambiguity that persists across all three models. These overlaps are not artifacts of insufficient feature engineering but reflect real behavioral similarity during specific traffic phases. Without additional discriminative signals, such as connection-level patterns or application identifiers, some degree of confusion between these class pairs is structurally unavoidable.

**Statistical inference without payload access.** The system operates entirely on aggregate flow statistics. It cannot distinguish two applications that produce statistically indistinguishable flow profiles, nor can it detect deliberate traffic shaping designed to mimic a different class's signature. Classification is probabilistic rather than deterministic, and its reliability degrades when application behaviour deviates from training-time patterns due to network conditions or software updates.

**Network and deployment generalizability.** Traffic characteristics are influenced by network path conditions, CDN configurations, server-side encoding settings, and client hardware. A model trained on captures from one environment may not transfer directly to substantially different network contexts without retraining or domain adaptation, limiting out-of-the-box deployment to conditions similar to those captured.

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## 7. Conclusion

This study demonstrates that seven categories of QUIC-encrypted application traffic, video streaming, live streaming, short-form video, file download, web browsing, chat, and VoIP, can be classified with

high accuracy using only flow-level metadata observable at the IP and UDP layers, without accessing any encrypted content. Three machine learning classifiers were trained and evaluated on a 296-record dataset constructed using a sliding time-window methodology applied to QUIC packet captures.

Random Forest and Decision Tree both achieved 91.67% overall accuracy, with macro-averaged F1-scores of 0.90 and 0.91 respectively. Random Forest demonstrated superior consistency across minority and boundary classes, particularly for short\_video (F1=0.89, recall=1.00) and live (precision=1.00), making it the recommended model for deployment. XGBoost achieved 83.33% accuracy, with performance constrained by the small sample sizes of underrepresented classes; its results are expected to improve substantially with a larger, more balanced dataset. Across all models, the video, download, and browsing classes were classified with near-perfect scores, while the short\_video/live and voip/chat class pairs represent the primary sources of misclassification, reflecting genuine behavioral overlap rather than model deficiency.

The findings confirm that QUIC's encryption model, while effective at protecting payload confidentiality, does not prevent traffic classification based on observable side-channel patterns in packet size distributions, inter-arrival timing, and burst behavior. Flow-level metadata classification achieves production-grade accuracy using features that are both technically and legally observable, establishing a viable framework for encrypted traffic management, quality-of-service enforcement, and network planning in QUIC-dominated environments.